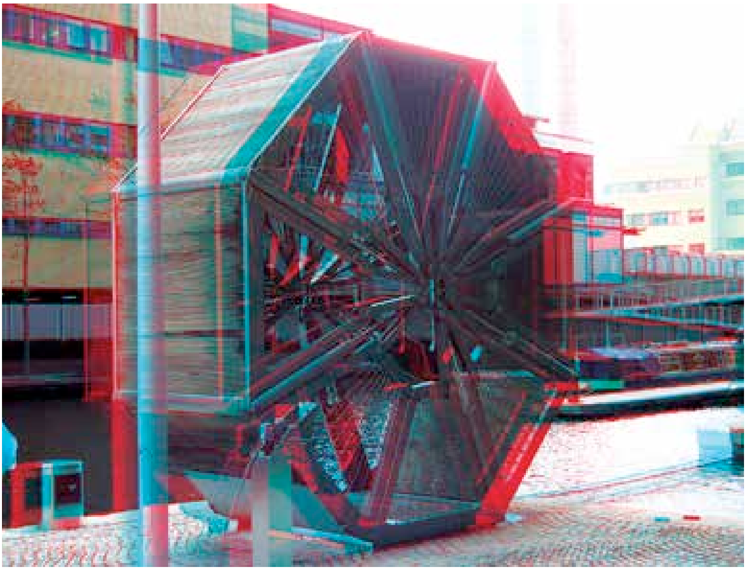
Perceiving Depth

Nature has built the human body based on needs. It is no different when it comes to our ability to perceive depth in the real-life around us. It is important to understand how close the attacking tiger is, it is essential to know whether you will survive the fall or not, or in modern day life, it is vital to know whether you have to leave your bean bag to get that remote or stretching out will suffice. A major part of our brain is dedicated to processing what our eyes see, each one of the pair, and using that to create a 3D reality in our head.



A stereoscopic image of a Rolling Bridge, which uses chromatic filtering to achieve 3D

The left eye perceives a certain two dimensional image of the surroundings. So does the right eye, but with a slightly different image. Due to the proximity of the pair of eyes, our brain processes the two images as one and overlays them to produce the sense of depth. 3D display technology is based on the same principle. Two slightly offset images are seen by each eye leading the brain to process them as a single image and perceive depth within them. Technically, this is not '3D', rather it is dual, or more accurately, stereoscopic 2D. Just like a combination of two stereo-speakers cannot be said to produce fully 3D audio, similarly so-called 3D nowadays is usually just dual-2D. But like many others, the misnomer has stuck.